

D1	1	SMAJ24CA	Bidirectional 400 W TVS; 24 V standoff, 26.7 V min breakdown, 38.9 V clamp	Diode_SMD:D_SMA	Bourns	SMAJ24CA			Bidirectional 24 V TVS candidate; 38.9 V specified clamp is pulse-condition dependent. Source energy, ringing and downstream peak voltage remain open.	Open datasheet
D3	1	STPST10H100SB	100V 10A Bidirectional 400 W TVS; 24 V standoff, 26.7 V min breakdown, 38.9 V clamp	PDB_Footprints:STPST10H100SB_DPAK	STMicroelectronics	STPST10H100SB-TR			Selected power clamp. Connect both anode leads to GND and cathode tab to PDB_VOUT. Pulse/thermal validation required.	Open datasheet
D4	1	SMAJ24CA	Bidirectional 24 V TVS candidate for local ESC connector ringing control; validate pulse energy and clamp voltage on the installed harness.	Diode_SMD:D_SMA	Bourns	SMAJ24CA			Selected: local bidirectional clamp from IDEAL_IN to U5 reference IDEAL_VSS.	Open datasheet
D5	1	SMBJ24CA	4 A, 32 VDC MINI blade fuse; backup protection for a fault the MOSFETs cannot interrupt	Diode_SMD:D_SMB	Littelfuse	SMBJ24CA			Procurement selection: Littelfuse SMBJ24CA. Retained for local ringing control; validate installed harness pulse energy and clamp behavior.	
F1	1	4A MINI		Fuse:Fuseholder_Blade_Mini_Keystone_3568	Littelfuse	0297004.WXNV	Keystone Electronics	3568	4 A 32 V MINI; load envelope 2.5 A maximum. Fuse, source and conductor coordination remains a release requirement.	Open datasheet
F2	1	SCHURTER UHS 100A 3-140-178	100 A surface-mount high-current fuse candidate for the dedicated ESC branch	PDB_Footprints:SCHURTER_UHS_8.4x9.4mm_PR OVISIONAL	SCHURTER	3-140-178			PROCUREMENT UPDATE: use 3-140-178 in place of original 3-140-177. SCHURTER lists identical 100 A / 50 V UHS electrical ratings and package dimensions; order-number difference is packaging. DigKey offers 3-140-178 cut tape. Engineering release checks still apply: verify 60 A continuous temperature rise, 100 A/30 s time-current behavior, 45 pack fault current, breaking capacity, and heavy-copper land pattern.	Open datasheet
H1	0	MountingHole	Mounting Hole without connection	MountingHole:MountingHole_3.2mm_M3_ISO1 4580						
H2	0	MountingHole	Mounting Hole without connection	MountingHole:MountingHole_3.2mm_M3_ISO1 4580						